Effect of healthy lifestyle behaviors on the association between leukocyte telomere length and coronary artery calcium.

The telomere length is an indicator of biologic aging, and shorter telomeres have been associated with coronary artery calcium (CAC), a validated indicator of coronary atherosclerosis. It is unclear, however, whether healthy lifestyle behaviors affect the relation between telomere length and CAC. In a sample of subjects aged 40 to 64 years with no previous diagnosis of coronary heart disease, stroke, diabetes mellitus, or cancer (n = 318), healthy lifestyle behaviors of greater fruit and vegetable consumption, lower meat consumption, exercise, being at a healthy weight, and the presence of social support were examined to determine whether they attenuated the association between a shorter telomere length and the presence of CAC. Logistic regression analyses controlling for age, gender, race/ethnicity, and Framingham risk score revealed that the relation between having shorter telomeres and the presence of CAC was attenuated in the presence of high social support, low meat consumption, and high fruit and vegetable consumption. Those with shorter telomeres and these characteristics were not significantly different from those with longer telomeres. Conversely, the subjects with shorter telomeres and less healthy lifestyles had a significantly increased risk of the presence of CAC: low fruit and vegetable consumption (odds ratio 3.30, 95% confidence interval 1.61 to 6.75), high meat consumption (odds ratio 3.33, 95% confidence interval 1.54 to 7.20), and low social support (odds ratio 2.58, 95% confidence interval 1.24 to 5.37). Stratification by gender yielded similar results for men; however, among women, only fruit and vegetable consumption attenuated the shorter telomere length and CAC relation. In conclusion, the results of the present study suggest that being involved in healthy lifestyle behaviors might attenuate the association between shorter telomere length and coronary atherosclerosis, as identified using CAC.